

The effects of aerobic- versus strength-training on body image among young women with pre-existing body image concerns



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ABSTRACT

This experiment compared the effects of aerobic-training (AT) versus strength-training (ST) on body image among young women with pre-existing body image concerns. Theory-based correlates of body image change were also examined. Participants were 46 women (M age = 21.5 years), randomly assigned to an 8-week AT or ST intervention consisting of supervised exercise 3 days/week. Multidimensional measures of body image were administered pre- and post-intervention, along with measures of physical fitness, perceived fitness, and exercise self-efficacy. Women in the AT condition reported greater reductions in social physique anxiety ($p = .001$) and tended to report greater improvements in appearance evaluation ($p = .06$) than women in the ST condition. Changes in perceived fatness, perceived aerobic endurance and aerobic self-efficacy were significantly correlated with body image change ($ps < .003$). Results provide direction for prescribing exercise to improve body image and advancing theory to account for the effects of exercise.

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Introduction

Rates of body dissatisfaction are high among women living in westernized societies (Cash, 2011). For example, an online survey (Frederick, Peplau, & Lever, 2006) of nearly 27,000 women revealed that only 41% felt “good” or “great” about their bodies, while the majority felt that their bodies were “just okay” or even unattractive. These statistics are cause for concern; poor body image is associated with low self-esteem, is believed to be a cause of anxiety and depression, and plays a significant role in eating disorder etiology (Polivy & Herman, 2002; Stice & Whitenton, 2002). Given the importance of body image to health and well-being, there is a need for evidence-based treatments to improve how women think and feel about their bodies.

Considerable research, including three meta-analyses (Campbell & Hausenblas, 2009; Hausenblas & Fallon, 2006; Reel et al., 2007), indicates that exercise-training programs are effective for improving women's body image. These meta-analyses, however, have produced conflicting results regarding the type of

exercise – strength training (e.g., lifting weights, calisthenics) or aerobic training (e.g., walking, cycling) – that is most effective. In two of the meta-analyses (Hausenblas & Fallon, 2006; Reel et al., 2007), strength-training interventions were grouped with other anaerobic activity interventions (e.g., stretching, sprinting, playing sports such as baseball or volleyball). One of these meta-analyses (Hausenblas & Fallon, 2006) reported no differences in experimental studies of the effects of anaerobic (Cohen's $d = 0.27$) versus aerobic ($d = 0.25$) exercise interventions on body image. Another, which collapsed body image measures along with measures of body esteem and physical self-concept (Reel et al., 2007), found larger effects for anaerobic ($d = 0.64$) than aerobic exercise ($d = 0.40$). A third, more recent meta-analysis found no difference in body image improvements for exercise interventions that employed strength training and other forms of resistance exercise ($d = 0.37$) versus interventions involving aerobic exercise ($d = 0.30$; Campbell & Hausenblas, 2009). In addition, one study (Tucker & Mortell, 1993) directly compared the effects of strength versus aerobic training on body image.¹ Using an experimental design, women between the ages of 35 and 49 were randomly assigned

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¹ We acknowledge that other studies have compared the effects of interventions that combine exercise modalities such as Henry, Anshel, and Michael's (2006) study that compared an aerobic intervention with a combined aerobic +

to an unsupervised, home-based 12-week exercise program consisting of 3 days/week of either strength training or walking. Although body image improved in both conditions, improvements were significantly greater in the strength-training condition. Taken together, given contradictory findings, it is impossible to conclude from the existing literature if there are differential effects of aerobic versus strength-training exercise on women's body image.

On the one hand, strength training may be more effective than aerobic exercise because each strength-training session has the potential to provide exercisers with immediate positive feedback about their capabilities (Martin Ginis, Eng, Arbour, Hartman, & Phillips, 2005). Unlike typical aerobic workouts (e.g., group fitness classes), a single strength-training session can provide the exerciser with immediate feedback on her progress and level of function (i.e., by virtue of knowing the amount of weight she can lift). Strength-training may also help women become more aware of their body's functional capabilities, which could lead to a de-emphasis on physical appearance (Martin Ginis et al., 2005). On the other hand, aerobic exercise may have stronger effects on body image given that women's body dissatisfaction is primarily driven by concerns about being overweight. Because women typically perceive aerobic exercise as being more conducive to weight loss than other types of exercise (Prichard & Tiggemann, 2008) they may become more satisfied with their bodies by virtue of engaging in a behavior that could allay concerns about being overweight.

Research that directly compares the effects of aerobic versus strength-training exercise on body image is important for both clinical and theoretical reasons. From a clinical perspective, if scientists can determine the type of exercise that is most conducive to improving women's body image, then interventionists can develop maximally effective exercise treatments. From a theoretical perspective, if there are differences in the extent to which different types of exercise improve body image, such knowledge could lead to a better understanding of the mechanisms that account for exercise-induced body image change. For instance, if strength training yields greater effects than aerobic exercise, then this might suggest that a mechanism associated with muscularity or strength gains contributes to exercise-induced body image change. If aerobic exercise is more effective, this finding might indicate that changes in body fat or aerobic fitness explain the effects of exercise.

Given the need for greater knowledge regarding the body image benefits of each type of exercise (Prichard & Tiggemann, 2008), the primary purpose of the present study was to compare the effects of an 8-week supervised program of aerobic exercise versus strength training on changes in body image among young women. In order to maximize the clinical relevance of this study, we recruited women with pre-existing body image concerns. It was predicted that both types of exercise would have significant effects on body image. In the absence of a strong empirical, conceptual, or theoretical rationale, an a priori hypothesis could not be formulated regarding the superiority of one type of exercise over the other.

A secondary study purpose was to examine the relationship between changes in body image and changes in variables that might account for the effects of exercise on body image. Research that advances an understanding of mechanisms and the development of theories to explain the effects of exercise on body image has been sorely lacking. In fact, there is currently no explicit model or theory to guide exercise and body image research (Martin Ginis, Bassett-Gunter, & Conlin, 2012). As such, some researchers (e.g., Lindwall & Lindgren, 2005; Martin Ginis, McEwan, Josse, & Phillips, 2012) have turned to the Exercise and Self-Esteem Model (EXSEM; Sonstroem

& Morgan, 1989) as a framework for designing studies and testing hypotheses pertaining to exercise and body image.

The EXSEM is a model designed to explain the effects of exercise on global self-esteem. It consists of hierarchically organized constructs that lead from changes in measures of physical performance and fitness to changes in self-esteem. According to the EXSEM, exercise-related changes in *actual physical fitness and/or performance* provide information that affects one's physical self-efficacy (i.e., self-efficacy to perform specific physical tasks such as running or lifting). *Physical self-efficacy* in turn, informs more generalized evaluations of one's *physical competence*, which are operationalized as perceptions of one's own strength, fitness, fatness and other aspects of the physical self (Sonstroem, Speliotis, & Fava, 1992). Physical self-perceptions can then influence self-esteem directly, or indirectly through *physical acceptance*, a construct that is operationalized as scores on measures of body image (Sonstroem et al., 1992). This latter operationalization facilitates the application of the EXSEM to studies examining the effects of exercise on body image. In such studies, the model has been used to generate hypotheses pertaining to the effects of changes in actual physical fitness/performance, physical self-efficacy, and physical self-perceptions (e.g., perceived physical fitness) on body image. Given the hierarchical ordering of the constructs, actual changes in physical fitness should have the most distal influence on body image, and changes in physical self-perceptions should have the most proximal influence. These relationships are depicted in Fig. 1, below the dotted line.

To date, only one published study has examined the role that the full set of EXSEM variables plays in accounting for changes in body image (Martin Ginis, McEwan et al., 2012). In that study, overweight women participated in a 16-week diet and exercise (combined aerobic and strength training) intervention. Of the three EXSEM constructs purported to influence body image, changes in perceived fitness (i.e., physical competence) – particularly changes in perceived body fat and perceived physical strength – accounted for most of the explained variance in body image change. When all of the variables were entered into a regression model to predict body image change, changes in aerobic self-efficacy, perceived body fat, and perceived strength were the only significant predictors at Week 8; changes in perceived body fat and perceived strength were the only significant predictors at Week 16. Actual physical fitness changes did not predict body image change at either time point.

These findings suggest that how an exerciser *perceives* changes to her body is a stronger determinant of body image change than the actual magnitude of those changes. A limitation of the study however, is that it is impossible to separate the effects of the exercise from the effects of the dietary intervention. Nevertheless, the results are consistent with a systematic review of exercise training studies which concluded that actual changes in physical fitness are only weakly related to body image change, whereas perceived changes in physical fitness and self-efficacy are consistently positive predictors of body image change (Martin Ginis, Bassett-Gunter et al., 2012). As such, we predicted that following an 8-week exercise intervention, body image change would be positively, significantly correlated with changes in perceived fitness and exercise self-efficacy, and unrelated to actual physical fitness changes.

Method

Participants

Based on meta-analytic findings (Hausenblas & Fallon, 2006) that women's exercise interventions incorporating strength or aerobic training have a large effect on body image ($d = 0.45$), $N = 46$ (23 participants/condition) was required to have 80% power to detect

strength-training + anaerobic exercise intervention. Combined interventions are not the focus of this study.

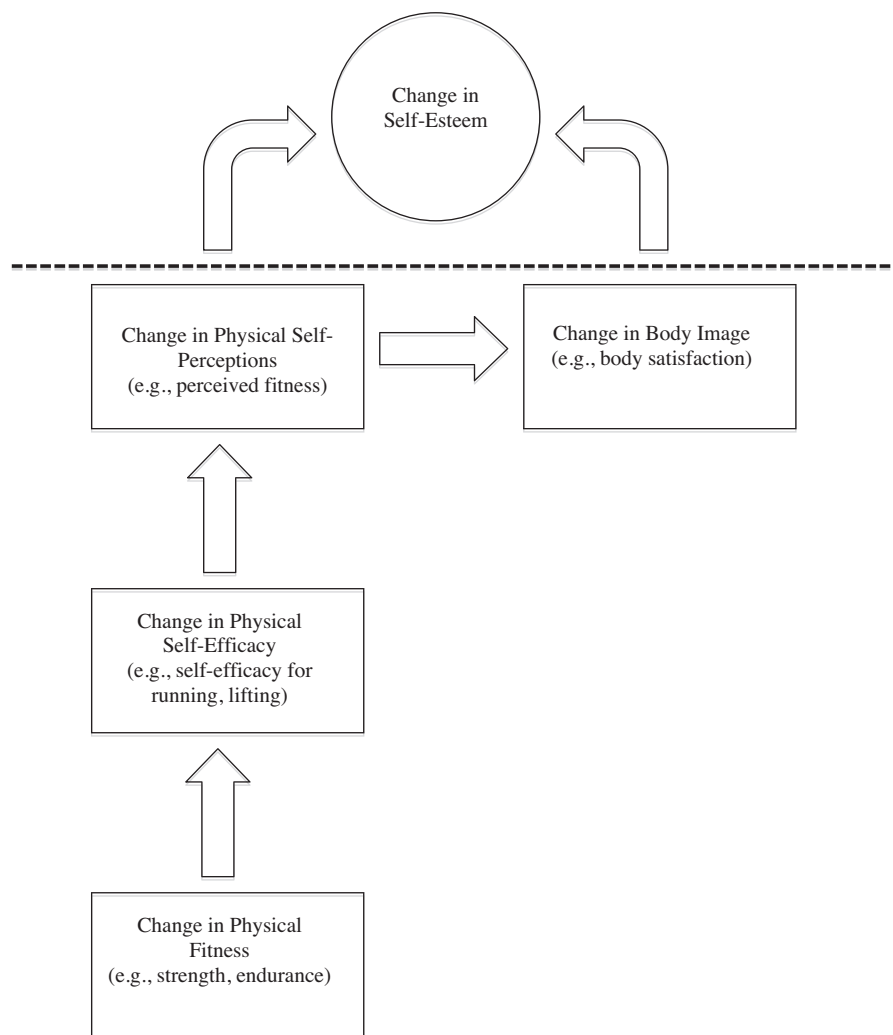


Fig. 1. The Exercise and Self-Esteem Model (Sonstroem & Morgan, 1989) as operationalized and applied for testing the effects of exercise on body image.

a similar main effect of exercise training over time ($\alpha < .05$; Cohen, 1992). Given the conflicting meta-analytic data, it was not possible to accurately determine the a priori sample size needed to detect significant differences in body image change across the two types of exercise.² As such, a decision was made to power the study to detect a main effect over time, and to calculate and interpret the magnitude of the Condition $\times$ Time interaction effect, to determine if the two types of exercise have differential effects on body image.

Participants were recruited from a university campus through advertisements targeting women who felt dissatisfied with their bodies and who wanted help following through with their New Year's resolutions to exercise. All participants were informed that the purpose of the study was "to examine the role of different physiological and psychological mechanisms underlying the exercise-body image relationship." Participants did not know that we were studying the differential effects of aerobic versus strength-training exercise. Body image improvement was not indicated as a specific study goal.

Study recruitment, testing, and training occurred between January 2007 and April 2007. Study inclusion criteria were:

(a) exercise $<$ twice/week and ≤ 30 min/session over the previous year; (b) pre-existing body image concerns, operationalized as scoring ≥ 27 on the 9-item Social Physique Anxiety Scale (SPAS; Martin, Rejeski, Leary, McAuley, & Bane, 1997) and ≤ 3 on the Body Areas Satisfaction Scale (BASS; Cash, 2000); and (c) no exercise contraindications. Relatively inactive women with poor body image were sampled as they may be most responsive to an exercise-based body image intervention (Martin Ginis & Bassett, 2011). Participants were offered \$75 in return for study completion. Eighty-two women expressed interest and were screened for study inclusion; 46 were eligible and enrolled ($M_{\text{age}} = 21.52$, $SD = 2.93$; ethnicity = 80% Caucasian; $M_{\text{BMI}} = 22.96$, $SD = 3.89$).

Measures

Body image. In recognition of the multidimensional nature of body image, the affective, cognitive, and subjective satisfaction dimensions (Stewart & Williamson, 2004) were assessed. The affective dimension was measured with the 9-item (Martin et al., 1997) Social Physique Anxiety Scale (SPAS; Hart, Leary, & Rejeski, 1989), which assesses anxiety experienced in response to others' evaluations of one's body. Each item (e.g., "when it comes to displaying my physique/figure to others, I am a shy person") was rated on a 5-point scale ranging from 1 (*not at all characteristic*) to 5 (*a great deal*). After reverse scoring two items, a mean SPAS score was calculated with higher scores reflecting greater social physique

² One might argue that the study be powered to detect a small between-groups difference (i.e., a small effect). The a priori sample size estimation for 80% power to detect a small mean difference ($p < .05$) is $N = 786$ (Cohen, 1992). Clearly, this would not have been a realistic sample size for our exercise training study.

anxiety. The SPAS has demonstrated strong psychometric properties in studies involving college-aged women (for a review see Martin Ginis, Lindwall, & Prapavessis, 2007). In the present study, internal consistency was $\alpha = .87$ at pre-training and $\alpha = .87$ at post-training.

The cognitive and subjective satisfaction dimensions were assessed with the 7-item Appearance Evaluation (AE) and the 9-item Body Areas Satisfaction (BASS) subscales respectively, drawn from the Multidimensional Body Self-Relations Questionnaire (MBSRQ; Cash, 2000). The AE assesses general thoughts of physical attractiveness (e.g., “I like my looks just the way they are;” “most people would consider me good looking”) and the BASS assesses satisfaction with nine specific aspects of one’s appearance (e.g., overall appearance, lower torso, weight). Higher scores reflect greater satisfaction. The AE and BASS have demonstrated strong psychometric properties in studies involving college-aged women (for a review, see Bane & McAuley, 1998). In the present study, the internal consistency for AE was adequate (pre $\alpha = .86$; post $\alpha = .86$), whereas the internal consistency for BASS varied across measurement points (pre $\alpha = .62$; post $\alpha = .73$). Consistent with previous exercise intervention research (Martin Ginis et al., 2005), three items were deleted from the BASS (satisfaction with hair, face, height), which improved the pre-test reliability (pre $\alpha = .69$; post $\alpha = .72$).

Fitness. In the exercise sciences, physical fitness is operationalized “as [a set of] measurable health and skill-related attributes that include cardiorespiratory fitness, muscular strength and endurance, body composition and flexibility, balance, agility, reaction time and power” (Garber et al., 2011). The present study incorporated measures of three of these attributes: body composition, aerobic fitness, and muscular strength. *Body composition* was indirectly assessed as body mass index (BMI) and waist and hip circumference. Height was measured using a tape measure and weight using a standard scale (Lifesource Precision Personal Health Scale); BMI was then calculated as kg/m^2 . A tape measure was used to measure the largest diameter of waist and hip circumference in cm (American College of Sports Medicine [ACSM], 2008); waist-to-hip ratio was then calculated by dividing waist circumference by hip circumference.

Aerobic fitness was assessed using the Astrand Rhythmic Submaximal Fitness Test (Astrand & Rodahl, 1986; Legge & Banister, 1986). Participants cycled on a stationary bicycle at a constant workload (50 rpm) for 7 min. Heart rate was measured every minute using a Polar Heart Rate monitor (Polar Electro Inc., New York), and steady state heart rate at minutes 6 and 7 were averaged and compared to published nomogram tables to determine an estimation of VO_2 max (ml/kg/min). Previous research has shown steady state to be strongly associated with VO_2 max (i.e., $r_s = .85-.90$; Fitchett, 1985).

Muscular strength was assessed by 10 repetition maximum (10-RM) tests, which yield the maximal weight (in pounds) a person can lift for 10 repetitions. The 10-RM was used rather than the traditional 1-RM because in beginner strength-trainers, it is considered a safer and more valid assessment of muscular strength (Baechle, Earle, & Wathen, 2000). Four exercises were tested: chest/bench press, shoulder press, seated row, and leg press. A 5-min warm-up on a stationary bicycle was followed by the four exercises with progressively heavier weights until the participant could only complete 10 repetitions. Participants had up to three attempts to achieve their maximum and rested for 2 min between sets to ensure adequate recovery.

Perceived fitness. Consistent with other body image studies guided by the EXSEM (e.g., Lindwall & Lindgren, 2005; Martin Ginis, McEwan et al., 2012), changes in physical self-perceptions were operationalized as changes in perceived physical fitness. Perceived fitness was assessed with the Perceived Body Fat, Perceived Strength, and Perceived Physical Endurance subscales of the

Physical Self-Description Questionnaire (PSDQ; Marsh, Richards, Johnson, Roche, & Tremayne, 1994). These subscales assess participants’ agreement with descriptive items regarding their bodies (e.g., “I have too much fat on my body,” “I am a physically strong person,” “I would do well in a test of physical endurance and stamina”) on a scale ranging from 1 (*false*) to 6 (*true*). Each subscale consists of six items, which are averaged to create three composite scores. Higher scores for each subscale indicate less perceived body fat, greater perceived strength, and greater perceived endurance, respectively. Research supports the validity and reliability of the three subscales when administered to female university students (Asci, 2003). Internal consistencies were adequate for all perceived fitness measures at pre- and post-test ($\alpha_s = .89-.93$).

Exercise self-efficacy. Aerobic and strength-training exercise self-efficacy items were developed based on measures of exercise task self-efficacy used in previous research (e.g., Rodgers & Gauvin, 1998). Questionnaire format was consistent with Bandura’s (1997) self-efficacy measurement guidelines. Specifically, for aerobic self-efficacy, participants were asked “How confident are you that you could run on a treadmill, at a moderate intensity (not exhausting), without stopping?” for each of the following durations: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, and 60 min. Responses were made on a scale ranging from 0% *not at all confident* to 100% *completely confident* and averaged across the 12 items. Strength-training exercise self-efficacy was assessed with four questionnaires assessing self-efficacy for four exercises. Participants were asked: “How confident are you that you could [bench press; shoulder press; perform a seated row; leg press]” increasing amounts of weight. Responses were made on the 0–100% scale described above and were averaged for each questionnaire. Internal consistency estimates were adequate for all self-efficacy measures at pre- and post-training ($\alpha_s = .89-.96$).

Procedure

The protocol was approved by the university’s Research Ethics Board. Interested volunteers were emailed an information packet and screening questionnaire. Those eligible and who provided consent were scheduled for two pre-training assessment sessions.

Pre-training assessment Session 1. Participants provided demographic information and completed the body image and perceived fitness measures. They were then given an orientation to the 8-week exercise program consisting of a tour of the exercise facilities, a demonstration of the various exercise machines (e.g., treadmill, elliptical, strength-training equipment) and instructions on how to log their exercise sessions. Next, participants performed the aerobic fitness test and completed the aerobic exercise self-efficacy questionnaire.

Pre-training assessment Session 2. Session 2 was completed within a week of Session 1. Participants performed the 10-RM tests and then completed the strength-training exercise self-efficacy scales. They were paid \$25 for completing the two assessment sessions.

Exercise training programs. Using a random numbers table, participants were randomized to either the aerobic training (AT) or strength training (ST) condition. Participants were prescribed exercise 3 days/week for 8 weeks, with the goal of completing 24 exercise sessions. Participants were required to complete a minimum of 20 sessions to obtain the \$50 remuneration. All exercise sessions were supervised by female certified personal trainers who monitored exercise protocol and prescription adherence and verified all entries in participants’ exercise log books. Workout attire was standardized; participants were required to wear shorts and t-shirts at the exercise sessions.

In the AT condition, exercise was performed on an elliptical exercise machine or a treadmill. Training consisted of a 5 min warm

Table 1

Means, standard deviations, and significance tests for measures of body image, fitness, perceived fitness, and self-efficacy.

Measure	Strength-training condition (n = 23)		Aerobic-training condition (n = 17)		F tests		
	Baseline	Week 8	Baseline	Week 8	Time	Condition	Time × condition
<i>Body image measures</i>							
Appearance evaluation	2.78 (0.71)	3.11 (0.77)	2.86 (0.56)	3.53 (0.62)	31.56**	1.61	3.70
Body areas satisfaction	2.73 (0.48)	3.26 (0.55)	2.88 (0.47)	3.48 (0.41)	61.61**	1.78	0.36
Social physique anxiety	33.19 (4.98)	29.00 (7.59)	32.83 (5.80)	25.00 (5.75)	80.78**	1.0	9.41**
<i>Fitness</i>							
Body mass index (kg/m ²)	23.13 (3.98)	23.20 (4.30)	22.68 (4.04)	22.62 (4.00)	0.00	0.09	0.09
Waist-to-hip ratio	0.76 (0.04)	0.74 (0.07)	0.76 (0.06)	0.75 (0.05)	3.30	0.12	0.24
VO ₂ max (ml/kg/min)	31.80 (6.22)	35.04 (7.15)	33.00 (5.94)	38.39 (7.53)	32.31**	1.27	2.01
10-RM bench press (lb)	31.30 (10.79)	46.74 (11.04)	24.85 (9.78)	28.82 (7.81)	57.47**	16.88**	20.06**
10-RM shoulder press (lb)	24.35 (13.25)	33.70 (10.79)	17.35 (6.87)	18.53 (8.80)	12.89*	13.36*	7.77*
10-RM seated row (lb)	51.09 (8.52)	59.35 (7.88)	46.47 (8.62)	47.35 (8.31)	16.86**	11.82**	10.98**
10-RM leg press (lb)	146.74 (47.33)	216.30 (43.75)	116.76 (50.53)	167.06 (63.67)	116.98**	6.65	3.02
<i>Perceived fitness</i>							
Perceived body fat	3.39 (1.66)	3.68 (1.74)	3.54 (1.38)	4.27 (1.31)	15.61**	0.59	2.95
Perceived strength	3.04 (0.97)	4.03 (0.86)	2.70 (1.22)	3.72 (1.03)	71.19**	1.18	0.01
Perceived endurance	2.25 (0.97)	2.93 (0.96)	1.93 (0.89)	3.57 (1.03)	58.58**	0.35	10.18*
<i>Self-efficacy</i>							
Aerobic self-efficacy	51.90 (27.77)	62.72 (18.91)	42.13 (20.12)	78.31 (16.65)	9.05**	0.22	17.20**
Bench press self-efficacy	63.04 (24.32)	78.04 (17.03)	44.17 (22.59)	50.00 (22.89)	6.93*	16.03**	1.33
Shoulder press self-efficacy	52.47 (25.59)	63.13 (21.02)	41.50 (28.84)	51.84 (21.12)	4.71*	3.27	0.01
Seated row self-efficacy	62.94 (10.42)	82.66 (18.98)	43.46 (20.91)	70.13 (35.62)	29.21**	7.95**	0.66
Leg press self-efficacy	66.94 (18.65)	69.94 (22.77)	44.70 (23.39)	53.19 (31.00)	4.02*	7.80*	0.97

* $p \leq .05$.** $p \leq .001$.

up, 30 min of activity at the target heart rate zone, and a 10 min cool down. Participants started exercising at 65% of their maximum heart rate (moderate intensity) at Week 1, and increased by 5% every two weeks to 80% of their maximum heart rate (heavy intensity), which they reached at Week 6. This prescription met ACSM guidelines for aerobic training (Haskell et al., 2007). Intensity was monitored by participants taking their pulse every 10 min. Exercise sessions took place at the campus fitness facility where both male and female exercisers were present.

In the ST condition, training consisted of a very light intensity 5 min warm-up on a bike or treadmill, approximately 30 min of strength training and a 10 min cool down on a bike or a treadmill. The warm-up and cool down were performed at a very light intensity (defined as <57% of maximum heart rate; cf., Garber et al., 2011) because aerobic exercise at this intensity has no effect on body image ($d = -0.04$ in experimental studies; Hausenblas & Fallon, 2006), thus preventing the aerobic warm-up/cool-down from confounding the effects of the strength-training.

In order to control for approximate training time between the two groups, the ST program was divided into six upper body exercises (lat pull downs, chest press, shoulder press, seated row, biceps curl, and triceps extensions) and five lower body and core exercises (leg press, hamstring curls, leg extensions, abdominal crunches, and lower back extensions) performed on rotating days (e.g., upper body on Monday, lower body Wednesday, upper body Friday, lower body Monday, etc.) using a combination of resistance machines and free weights. Participants began exercising at 70% of their 10-RM (moderate intensity) and progressed to 80–90% (heavy intensity) by the end of the eight weeks. During Weeks 1–2, they performed eight repetitions and two sets/exercise. From Weeks 3–8, they performed 10–12 repetitions and three sets/exercise. The prescribed amount of weight for each exercise was increased (5–10 lbs for lower body exercises and 10–20 lbs for upper body exercises) for each participant when she was able to successfully complete the repetition range for all sets in the current prescription. Weekly volume of strength training was in line with ACSM guidelines (Haskell et al., 2007), and weekly time spent exercising was equivalent across the AT and ST conditions. Exercise sessions took place in an exercise

research training center where both male and female exercisers were present.

Post-training assessments. Upon completion of the 8-week training, participants repeated the measures described in the pre-training assessments section, in the same order, across two sessions. Participants were also interviewed to determine whether they had done any additional physical activity, or other types of exercise beyond that prescribed. Upon completion of the post-testing sessions, participants were paid \$50 and debriefed.

Statistical analyses. All analyses were conducted using SPSS v. 19. Changes in the body image variables were tested using separate general linear model (GLM) 2 (condition) × 2 (time) repeated measures analyses of variance (ANOVA). Effect sizes for between-groups differences were calculated using Cohen's d . Using Cohen's conventions, d s of 0.20, 0.50, and 0.80 were considered indicative of small, medium and large-sized effects, respectively. For descriptive purposes only, changes in the EXSEM variables were also tested using GLM 2 (condition) × 2 (time) repeated measures ANOVAs. To examine the association between change in body image and change in the EXSEM variables, residualized change scores were computed for the exercise self-efficacy and perceived fitness variables, and simple change scores were computed for the fitness measures. Pearson correlations were then computed between the change scores.

Results

Participant Retention and Preliminary Analyses

Of the 46 women enrolled in the study, two withdrew before baseline testing. Forty-four completed baseline testing and were randomized. Three women withdrew before training began and one withdrew at Week 4. Given constraints on the study timeline (i.e., training and testing could not extend beyond semester's end when student participants would leave campus), dropouts were not replaced. The final sample consisted of $n = 23$ in the ST condition and $n = 17$ in the AT condition. On average, participants completed 21.5 ($SD = 1.5$) out of 24 prescribed exercise sessions (i.e., 90%)

and adherence did not differ between groups ($p = .55$). None of the women reported activity that exceeded, or was different from, the prescribed exercise.

Data were screened for entry errors, missing data, and deviations from normality (Tabachnick & Fidell, 2007). There were no significant differences between conditions for the demographic variables (all $ps > .05$), or baseline assessments of body image, anthropometrics, or aerobic fitness (all $ps > .05$). However, women in the ST condition lifted more weight than women in the AT condition ($p < .05$) at the pre-test for bench press, shoulder press, and leg press exercises. As preliminary analyses indicated these variables were not significantly correlated with any other variable, the baseline differences were not considered problematic and they were not included as covariates in any analyses.

Effects of Exercise Training on Body Image

Table 1 presents the ANOVA results for each body image measure. For the SPAS, as predicted, a significant main effect for time ($p < .0001$) indicated reductions in social physique anxiety across both conditions. The main effect was superseded by a significant Condition $\times$ Time interaction, $F(1, 39) = 9.41$, $p = .004$, partial $\eta^2 = .20$. Effect sizes for the decreases in social physique anxiety were very large for the AT condition ($d = -1.36$) and medium-sized for the ST condition ($d = -0.65$). For the AE subscale, as hypothesized, there was a significant main effect for time ($p < .001$) indicating that women in both conditions had significant improvements in thoughts about their bodies. The Condition $\times$ Time interaction effect did not achieve conventional levels of significance, although the effect itself was medium-sized (Cohen, 1992), $F(1, 39) = 3.70$, $p = .06$, partial $\eta^2 = .09$. The effect size for change in appearance evaluation for women in the AT condition was more than double ($d = 1.13$) the effect size observed for women in the ST condition ($d = 0.45$). For the BASS subscale, a significant main effect for time ($p < .001$) indicated that women in both conditions had a significant increase in body satisfaction from pre- to post-training. The Condition $\times$ Time interaction effect was not significant, $F(1, 39) = 0.36$, $p = .55$, partial $\eta^2 = .01$.

Effects of Training on Fitness, Perceived Fitness, and Exercise Self-Efficacy

Before examining the correlations between changes in body image and changes in the EXSEM variables, it was considered prudent to determine whether changes had indeed occurred in the EXSEM variables. The results are presented in Table 1 and summarized below.

Fitness. There were no significant main effects or interactions for BMI or waist-to-hip ratio ($ps > .08$). For aerobic fitness, the only significant effect was for time, with both conditions increasing their VO_2 max over the training period ($p < .001$). For all four strength measures, significant main effects for condition and time emerged ($ps < .05$). These main effects were superseded by significant Condition $\times$ Time interactions for the bench press, shoulder press, and seated row ($ps < .01$; partial $\eta^2 = .35$, .17, and .22, respectively); the ST condition demonstrated bigger increases in strength than the AT condition.

Perceived fitness. Main effects for time, but not condition, emerged for all three perceived fitness measures ($ps < .001$), indicating that women in both conditions experienced improvements in perceptions of their body fat, strength, and endurance. The only significant Condition $\times$ Time interaction was for perceived endurance, $F(1, 39) = 10.18$, $p = .003$, partial $\eta^2 = .21$; women in the AT condition perceived greater improvements in their aerobic endurance than women in the ST condition.

Exercise self-efficacy. There were main effects of condition on bench press, seated row, and leg press self-efficacy ($ps < .05$); overall, women in the ST condition scored higher on these measures. Significant main effects for time ($ps < .05$) emerged for all self-efficacy measures, indicating that in both conditions, training increased self-efficacy for jogging on a treadmill and the strength-training exercises. The only significant Condition $\times$ Time interaction was for aerobic self-efficacy, $F(1, 39) = 17.20$, $p < .001$, partial $\eta^2 = .31$; the AT condition reported greater increases than the ST condition.

Correlations between Changes in Measures of Body Image and Potential Mechanisms

Bivariate correlations between change in body image and change in the EXSEM variables are shown in Table 2. Improvements in all three body image measures were significantly correlated with changes in perceived body fat and perceived aerobic endurance (all $ps < .001$). Thus, women who had the greatest improvements in body image also reported the biggest perceived decreases in body fat and the biggest perceived increases in endurance. In addition, improvements in appearance evaluation and social physique anxiety were significantly correlated with increases in aerobic self-efficacy ($ps \leq .003$). Women who improved most on these two measures of body image also had the greatest increases in their self-efficacy to run without stopping for prolonged periods. No other correlations were statistically significant.³

Discussion

This experiment compared the effects of 8 weeks of aerobic versus strength-training exercise on body image among young women with pre-existing body image concerns. As hypothesized, both types of training led to significant improvements on all three body image measures. Furthermore, aerobic exercise yielded significantly greater improvements in social physique anxiety than did strength training and tended to produce greater improvements in appearance evaluation. (The effect for appearance evaluation did not achieve conventional levels of significance, although the effect itself was medium-sized.) In partial support of our secondary hypotheses and the tenets of the Exercise and Self-Esteem Model (EXSEM; Sonstroem & Morgan, 1989), improvements in all three measures of body image were significantly correlated with perceived changes in body fat and aerobic endurance; improvements in two of the body image measures were associated with an enhanced sense of aerobic self-efficacy. Actual changes in fitness were not correlated with change in any of the body image measures. Taken together, these results provide important new information regarding the relative effectiveness of aerobic and strength training for improving body image, as well as the utility of EXSEM variables for explaining such improvements.

The effect sizes for changes in appearance evaluation and social physique anxiety were twice as large for women in the aerobic condition versus the strength-training condition. These results conflict with Tucker and Mortell's (1993) finding that strength training was more effective than aerobic exercise (walking) for improving women's body image. As in the present study,

³ For exploratory purposes, the correlational analyses were also computed separately for the AT and ST conditions. The pattern of results was generally consistent across the two conditions; however, the relatively small ns limited our power to detect statistically significant correlations, particularly in the AT condition ($n = 17$). Because statistical significance differed across conditions, in the interest of parsimony, we have elected to present only the aggregated correlational data. Correlational matrices for the AT and ST conditions are available from the first author.

Table 2Bivariate correlations among change scores[†] for all study variables.

Changes in	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1. Appearance evaluation	1.00	.72**	-.56**	.48**	.08	-.19	-.01	-.28	.63**	.10	.56**	-.18	.25	-.19	-.13	.03	-.24	-.11
2. Body areas satisfaction		1.00	-.60**	.30	.01	-.23	-.16	-.23	.59**	.20	.49**	-.08	.06	-.17	-.06	-.07	-.31	-.04
3. Social physique anxiety			1.00	-.43**	.02	.23	.06	.20	-.57**	-.21	-.57**	.04	-.05	-.08	.20	-.15	.23	-.07
4. Aerobic self-efficacy				1.00	-.04	.13	-.13	-.16	.42**	-.09	.65**	-.02	.25	-.08	-.44**	-.32*	-.26	.04
5. Bench press self-efficacy					1.00	.72**	.72**	.32	.04	.16	-.05	.15	.24	-.07	.27	.45**	.26	.01
6. Seated row self-efficacy						1.00	.63**	.56**	-.19	.15	-.11	.21	.38*	-.13	.30	.24	.42**	-.01
7. Shoulder press self-efficacy							1.00	.25	-.17	.02	-.11	.10	.33*	.22	.09	.44**	.26	.01
8. Leg press self-efficacy								1.00	-.19	.40*	-.22	.23	.12	.18	.32*	.17	.34*	.03
9. Perceived body fat									1.00	.21	.55**	-.19	.04	-.19	-.16	-.03	-.32*	-.12
10. Perceived strength										1.00	.001	.08	.05	.03	.16	.24	-.18	-.01
11. Perceived endurance											1.00	.06	.24	-.08	-.21	-.15	-.01	-.01
12. Body mass index												1.00	-.13	.27	.25	.15	.42**	-.04
13. Waist:hip ratio													1.00	-.23	.13	.11	.17	.16
14. VO ₂ max (predicted)														1.00	-.04	.30	.22	.21
15. 10-RM bench press															1.00	.40*	.59**	.24
16. 10-RM shoulder press																1.00	.50**	.22
17. 10-RM seated row																	1.00	.17
18. 10-RM leg press																		1.00

* $p \leq .05$.** $p \leq .003$.[†] For the body image, self-efficacy, and self-perceptions variables, residualized change scores were used as the unit of analysis. For the fitness variables, simple change scores were used as the unit of analysis.

Tucker and Mortell's aerobic exercisers were prescribed moderate-heavy intensity activity. However, the implementation of that prescription differed considerably in our study versus theirs. Our participants engaged in supervised jogging on a treadmill and/or an elliptical trainer, whereas their participants walked, unsupervised. We suspect that by virtue of the supervision and the exercise equipment used (treadmills, elliptical trainers), our participants exercised at higher intensities, more consistently. As higher intensity exercise is key to eliciting body image change (Campbell & Hausenblas, 2009; Martin Ginis & Bassett, 2011), differences in intensity could explain our discrepant findings. Our results also diverge from a meta-analysis showing no differences between the effects of aerobic and strength training on body image (Campbell & Hausenblas, 2009). Given limitations of meta-analysis (i.e., studies with different methodologies and participant characteristics are combined), these differences are not surprising and attest to the importance of using experimental approaches to answer questions regarding the relative effectiveness of different exercise types for inducing body image change.

The EXSEM (Sonstroem & Morgan, 1989) was used as a framework to examine proposed mechanistic variables that might change along with body image. All of the EXSEM variables improved in both conditions, except for body composition. However, as predicted, only perceived fitness changes and exercise self-efficacy changes were correlated with body image change. These findings are consistent with the tenets of the EXSEM and a recent systematic review, showing that changes in perceived fitness and physical capabilities (i.e., self-efficacy) are more important than actual fitness changes for inducing body image change (Martin Ginis, Bassett-Gunter et al., 2012). These findings are also virtually identical to the results of an EXSEM-based study of predictors of body image change in overweight women participating in a diet and exercise weight-loss intervention (Martin Ginis, McEwan et al., 2012). Together, our results support the utility of the EXSEM as a framework for studying the effects of exercise on body image.

It is noteworthy that of the various perceived fitness and self-efficacy measures included in this study, changes in perceived body fat, aerobic endurance, and aerobic self-efficacy were the only significant correlates of body image change. The significant correlations observed between changes in body image and perceived body fatness attest to the importance of *feeling* thinner for evoking exercise-related body image change. This observation is not

particularly surprising given the centrality of thinness to the female body ideal. Yet, as our findings indicate, exercise-related changes in perceived fatness can occur without significant, objective changes in body shape. One possible explanation for this phenomenon is that the well-established, positive effects of exercise on overall mood and feeling states (e.g., Ekkekakis, 2013) have trickle-down effects on specific feeling states such as the experience of “feeling fat.” In other words, when exercise training leaves women feeling happier and more positive in general, they may, in turn, also experience their bodies more positively. Another possibility is that exercise diverts attention away from how the body looks, to how the body functions. For instance, a sedentary woman may think of her thighs as “fat.” However, as she lifts increasing loads on a leg-press machine, or runs increasingly long distances on a treadmill, she may become increasingly aware of the strength and power in her legs. Such awareness could counter perceptions of her thighs as “fat.”

Contrary to previous studies of correlates of exercise-related body image change in young women (Martin Ginis et al., 2005; Martin Ginis, McEwan et al., 2012), perceived strength changes were not significantly correlated with body image change. Rather, perceived aerobic endurance and aerobic self-efficacy were the only other significant correlates. Discrepant results may have emerged because unlike the previous studies, our sample was recruited based on pre-existing body image concerns, which were likely driven by concerns about thinness/fatness, rather than strength or muscularity. Thus, changes in perceived strength played a nonsignificant role in body image change because such perceptions were not particularly important to our participants' body image. In contrast, the significant correlations with perceived aerobic endurance and self-efficacy may reflect participants' underlying beliefs that aerobic exercise will help them move closer to their body ideal (Prichard & Tiggemann, 2008). Having greater perceived ability and confidence to engage in this type of activity may have led women to have more favorable thoughts and feelings about their bodies. It is also noteworthy that the aerobic training condition had significantly greater increases in perceived aerobic endurance and self-efficacy than the strength-training condition. Such differences may at least partially account for our finding that aerobic training had nearly double the effects of strength training on the outcome measures of social physique anxiety and appearance evaluation.

Limitations and Implications

Several study limitations warrant mention. First, the study was not adequately powered to detect significant interactions on all three body image measures. Given the relatively small sample size, we were also unable to test whether EXSEM variables functioned as statistical mediators of the effects of exercise on body image. Additional research is needed using appropriately powered designs, to further test perceived fatness, perceived aerobic endurance, and aerobic self-efficacy as mechanisms underlying the effects of exercise on body image. A second limitation is that we relied on the EXSEM to guide the selection of correlates of body image change. Other variables not captured by this model could be important. For example, Fox (2000) has proposed an undetermined psychophysiological mechanism that enhances mood and positive self-regard. A third limitation is that because of logistical constraints, the AE and ST conditions exercised in different settings. It might be argued that the greater reductions in social physique anxiety observed in the AE condition were a consequence of exposure effects, as these participants became accustomed to exercising in a public setting with high evaluation potential. This explanation seems unlikely, however, given that both conditions exercised in busy, mixed-sex gym settings. Furthermore, the aerobic exercise equipment utilized by the AE condition was deliberately situated in a separate mezzanine level of the fitness center, in order to minimize the conspicuousness of equipment users. Another limitation is that our measures of body composition were relatively simple. More comprehensive measures (e.g., DXA assessments of fat- and fat-free body mass) may have yielded different results. We also acknowledge that because testing our research question required a study design with separate aerobic and strength-training conditions, the external validity of the latter condition is questionable given that few women engage exclusively in strength training. Finally, although time spent exercising was held constant across the two conditions, absolute volume of exercise (i.e., kcal expended) was not necessarily equal. Further research is needed to determine if aerobic exercise remains a more effective treatment when compared to strength-training programs of equal metabolic demand.

Despite these limitations, our experimental investigation of the effects of supervised strength- and aerobic-exercise, prescribed and delivered in accord with ACSM exercise guidelines, makes at least two significant contributions to the body image literature. First, our study has shown that among young women with pre-existing body image concerns, both aerobic and strength training are effective for improving scores on measures of social physique anxiety, appearance evaluation and body areas satisfaction. However, for at least one of these measures, aerobic exercise may produce greater improvements than strength training. This information is invaluable for clinicians and other interventionists who prescribe exercise as a body image treatment. Second, our study contributes to the use and development of much-needed theories to guide exercise and body image research and intervention. Specifically, our results confirm hypotheses driven by the Exercise and Self-Esteem Model, and highlight the importance of changes in perceived body fat, aerobic endurance, and aerobic self-efficacy as potential mechanisms underlying exercise-induced body image improvements.

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